

How seeds are spread

AS ALL GARDENERS KNOW, a patch of bare soil never stays bare for long. Within days, seedlings start to spring up, and if the conditions are right, they eventually cover the ground. Even if the soil is sterilized by heating, so that all the seeds are killed, more somehow arrive and germinate. Plants have evolved some very effective ways of spreading their seeds. In certain plants, exploding seed pods fling the seeds into the air. Others have flying or floating seeds or fruits that are carried far and wide by the wind and by water currents. Animals also play their part. Many plants have fruits with hooks that stick to fur. The seeds of some species develop inside tasty berries. The berries are eaten by animals and birds, but the seeds pass through the digestive system of these creatures unharmed and fall to the ground, where they germinate.

Fruits have hooks

Each fruit has many tiny hooks

Lesser burdock

HITCHING A RIDE

The best way to find out which seeds are dispersed by animals is to go for a walk through rough grassland. You will probably return home with the fruits of a number of different plants stuck to your clothes. Known as burrs, these fruits have hooks and spines that cling to the fur and coats of passing animals. When the burrs are rubbed or scratched off, their seeds fall to the ground and germinate.



Burrs clinging to the fur on a dog's back

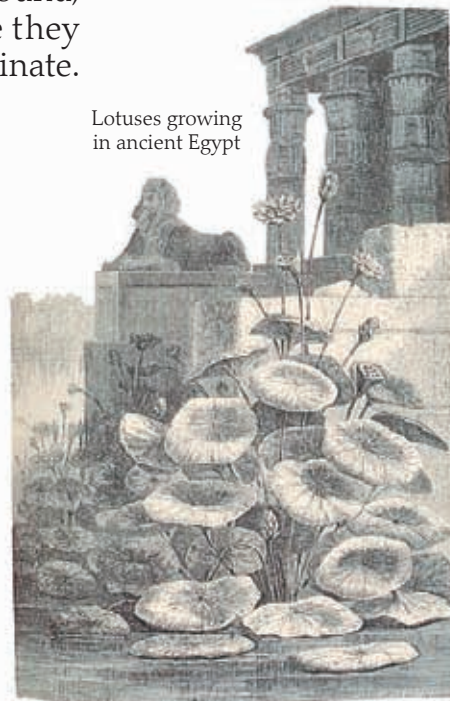


Lotus seed heads

Seed held in a cup

Dried lotus seed head, seen from above

Lotuses growing in ancient Egypt



WASHED AWAY

The lotus is a water plant that produces its seeds in a flattened head. When the seeds are ripe, they fall on to the water's surface and float away. Lotus seeds can live for an extraordinarily long time. Some have been known to germinate more than 200 years after they were shed.